**SAP ID: 60003240262**

**COURSE CODE: DJS23ILPC501** **DATE: 15/09/2026**

**COURSE NAME: Computer Networks Laboratory** **CLASS: T. Y. B.Tech**

**Experiment No.7**

**CO/LO:** Describe the fundamental components of computer networks and layered reference models.

**Aim:** Use Wireshark to understand the operation of TCP/IP layers :

- Ethernet Layer : Frame header, Frame size etc.
- Data Link Layer : MAC address, ARP (IP and MAC address binding)
- Network Layer : IP Packet (header, fragmentation), ICMP (Query and Echo)
- Transport Layer: TCP Ports, TCP handshake segments etc.
- Application Layer: DHCP, FTP, HTTP header formats

**Theory:**

Wireshark, a network analysis tool formerly known as Ethereal, captures packets in real time and displays them in human-readable format. Wireshark includes filters, color-coding and other features that let you dig deep into network traffic and inspect individual packets.

**Applications:**

- Network administrators use it to troubleshoot network problems
- Network security engineers use it to examine security problems
- Developers use it to debug protocol implementations
- People use it to learn network protocol internals beside these examples can be helpful in many other situations too.

**Features:**

The following are some of the many features wireshark provides:

- Available for UNIX and Windows.
- Capture live packet data from a network interface.
- Open files containing packet data captured with tcpdump/WinDump, Wireshark, and a number of other packet capture programs.
- Import packets from text files containing hex dumps of packet data.

- Display packets with very detailed protocol information.
- Export some or all packets in a number of capture file formats.
- Filter packets on many criteria.
- Search for packets on many criteria.
- Colorize packet display based on filters.● Create various statistics.

**Capturing Packets**

After downloading and installing wireshark, you can launch it and click the name of an interface under Interface List to start capturing packets on that interface. For example, if you want to capture traffic on the wireless network, click your wireless interface. You can configure advanced features by clicking Capture Options.

As soon as you click the interface's name, you'll see the packets start to appear in real time. Wireshark captures each packet sent to or from your system. If you're capturing on a wireless interface and have promiscuous mode enabled in your capture options, you'll also see the other packets on the network.

Click the stop capture button near the top left corner of the window when you want to stop capturing traffic.

Wireshark uses colors to help you identify the types of traffic at a glance. By default, green is TCP traffic, dark blue is DNS traffic, light blue is UDP traffic, and black identifies TCP packets with problems — for example, they could have been delivered out-of-order.

**Filtering Packets**

If you're trying to inspect something specific, such as the traffic a program sends when phoning home, it helps to close down all other applications using the network so you can narrow down the traffic. Still, you'll likely have a large amount of packets to sift through. That's where Wireshark's filters come in.

The most basic way to apply a filter is by typing it into the filter box at the top of the window and clicking Apply (or pressing Enter). For example, type “dns” and you'll see only DNS packets. When you start typing, Wireshark will help you autocomplete your filter.

Another interesting thing you can do is right-click a packet and select Follow TCP Stream. You'll see the full conversation between the client and the server. Close the window and you'll find a filter has been applied automatically — Wireshark is showing you the packets that make up the conversation.

**Inspecting Packets**

Click a packet to select it and you can dig down to view its details. You can also create filters from here — just right-click one of the details and use the Apply as Filter submenu to create a filter based on it.

Wireshark is an extremely powerful tool, and this tutorial is just scratching the surface of what you can do with it. Professionals use it to debug network protocol implementations, examine security problems and inspect network protocol internals.

**TCP Three-way Handshake**

The delta value between frames 1 and 2 can be used as a TCP transport connect baseline value. Other important information gathered from this handshake: • Window Size • SACK • Maximum Segment Size • Window Scale Option value

**Output (Wireshark capture on the lab machine):**

Packets were captured live on the Ethernet interface using Wireshark and then analysed with display filters and the packet details pane.

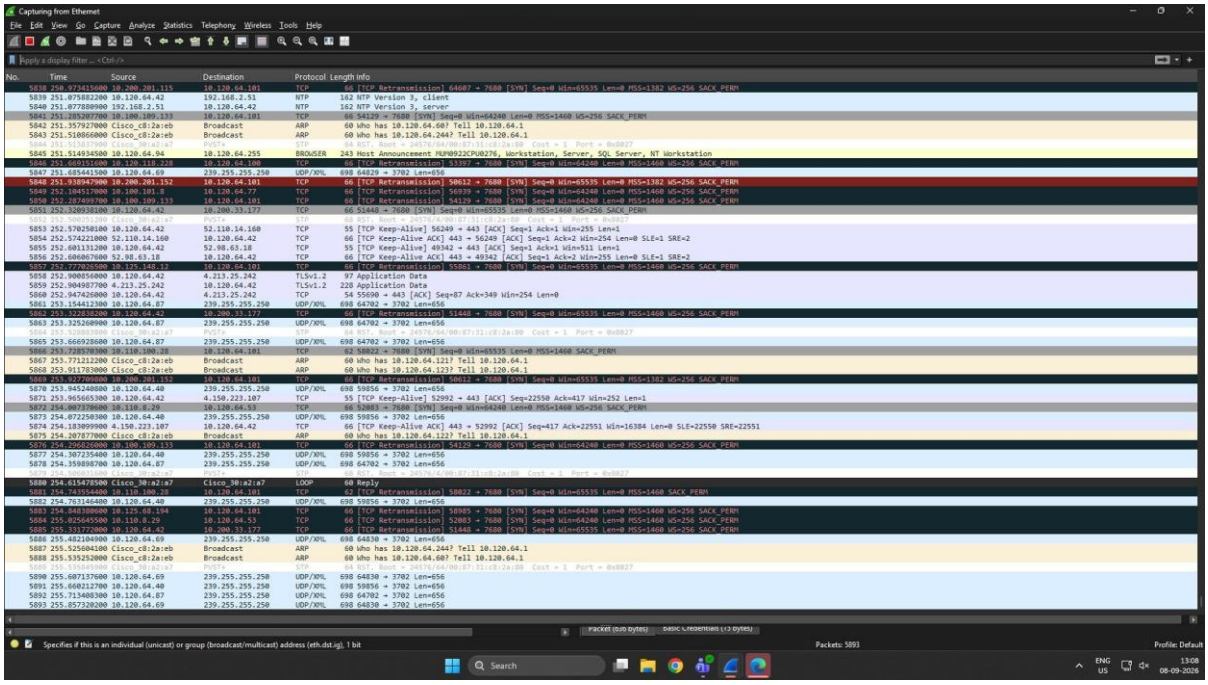


*Fig. 1: Live packet capture on the Ethernet interface (5893 packets captured): TCP, ARP, NTP, STP, TLS and UDP traffic, colour-coded by protocol.*

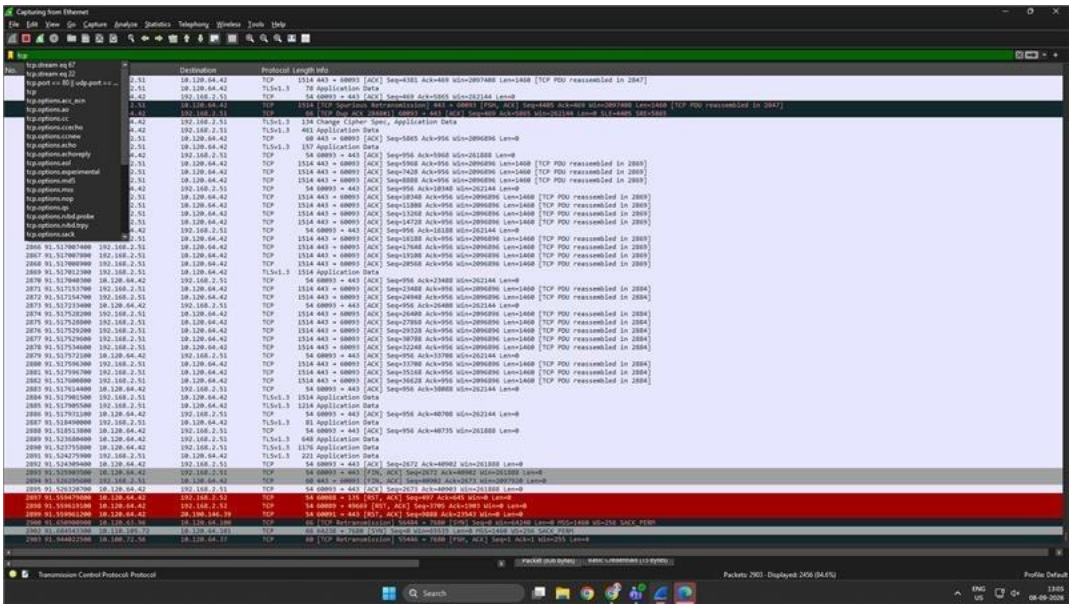


*Fig. 2: Display filter "tcp" applied: 2456 of the 2903 captured packets (84.6%) are displayed.*

## Department of Information Technology

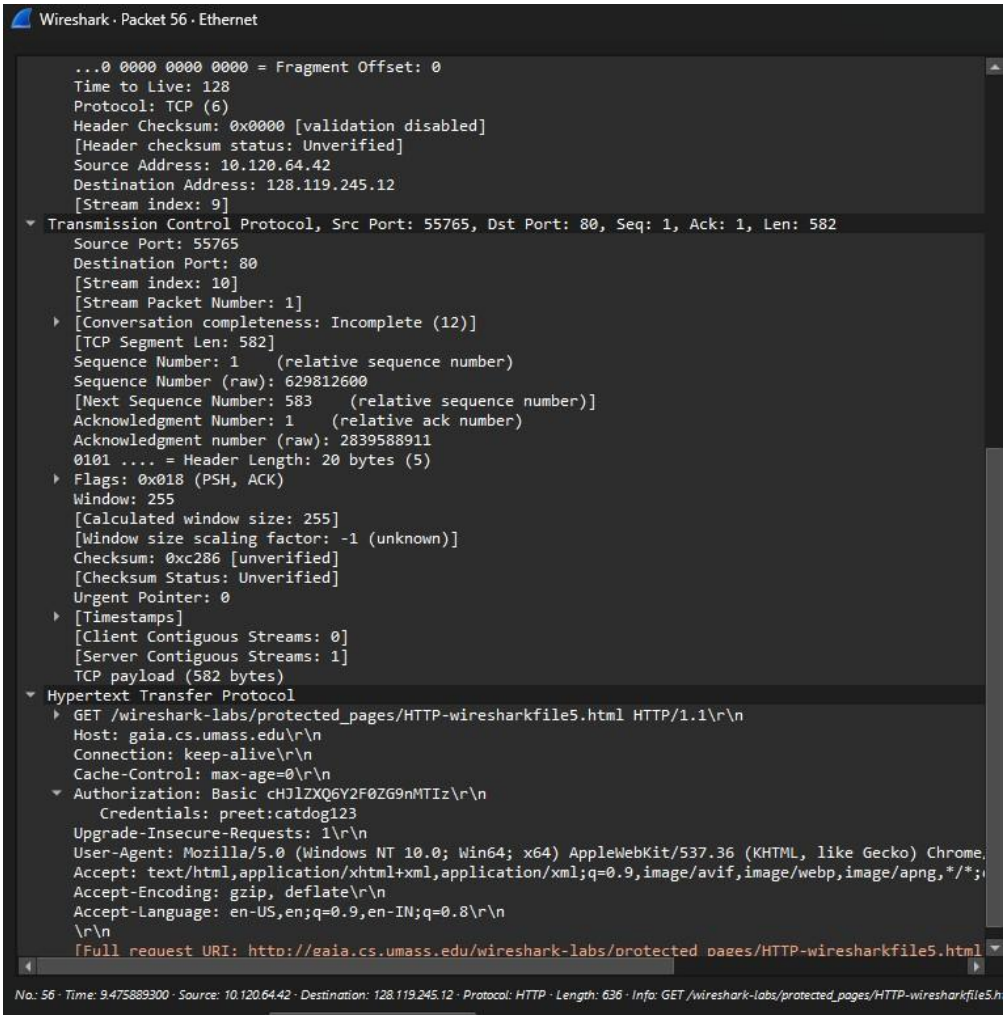



*Fig. 3: Packet 56 (HTTP GET request, 636 bytes) opened in a separate window, showing its IP, TCP and HTTP headers.*

- **Network layer (IP):** source 10.120.64.42, destination 128.119.245.12, TTL 128, protocol TCP (6).
- **Transport layer (TCP):** source port 55765, destination port 80, flags PSH/ACK (0x018), window 255, payload 582 bytes.
- **Application layer (HTTP):** GET /wireshark-labs/protected_pages/HTTPwiresharkfile5.html HTTP/1.1 with Host and Authorization (Basic) headers.

**Conclusion:** Studied and implemented packet sniffer tools like wireshark, ethereal, tcpdump to capture packet, analyse the packets captured by Wireshark.